

AI-Driven Protein Design with Diffusion and Language Models: A Comprehensive Survey

Abstract

The computational design of novel proteins with tailored structural and functional properties has historically been limited by the combinatorial explosion of the amino acid sequence space and the intricate physics governing macromolecular folding. In recent years, deep generative artificial intelligence has catalyzed a paradigm shift, transitioning the field from physics-based energy function optimization to data-driven probabilistic modeling. This survey provides an exhaustive review of the contemporary landscape of AI-driven protein design, focusing on two dominant generative frameworks: Protein Language Models (PLMs) and Diffusion Models. By systematically dissecting the architectural and mathematical foundations of these models—including Masked Language Models (MLMs), Autoregressive Models, Continuous Structure Diffusion, Discrete Sequence Diffusion, and Flow Matching—this report elucidates how generative AI is solving complex challenges such as inverse folding, motif scaffolding, and multiprotein complex design. Furthermore, this survey explores the integration of Generative Flow Networks (GFlowNets), Bayesian Optimization, and active learning frameworks that bridge the gap between computational generation and *in vitro* experimental validation. Through a synthesis of 300 recent scholarly publications, this manuscript details the state-of-the-art methodologies, highlights critical bottlenecks such as out-of-distribution generalization and dataset bias, and projects future trajectories toward fully programmable, autonomous protein engineering.

Introduction

Proteins are the fundamental molecular machines of biology, orchestrating structural, catalytic, and regulatory functions within all living systems. The primary amino acid sequence of a protein dictates its three-dimensional topological fold, which subsequently determines its biological function—a core principle formalized by Anfinsen’s thermodynamic hypothesis [33]¹. The primary objective of protein engineering is to navigate this sequence-structure-function paradigm to discover or design novel proteins for therapeutics, synthetic biology, and industrial biocatalysis. However, natural evolution has sampled only an infinitesimal fraction of the theoretical sequence space. For a relatively small protein of 100 amino acids, the number of possible sequences is 20^{100} , an astronomical expanse that renders brute-force combinatorial search mathematically impossible [24, 25]².

Historically, computational protein design was dominated by physics-based heuristic sampling algorithms. Platforms such as Rosetta utilized handcrafted energy functions, combining molecular mechanics with statistical potentials to evaluate the thermodynamic stability of

candidate sequences threaded onto target backbones [23, 32]². While physics-based rational design achieved notable milestones—including the design of novel enzymes and viral inhibitors—it was frequently hampered by high computational costs, reliance on rigid-backbone approximations, and limited experimental success rates that necessitated massive, high-throughput wet-lab screening [63, 67, 82]⁴.

The advent of deep learning has fundamentally rewritten the rules of protein design. Following the watershed success of AlphaFold2 and RoseTTAFold in accurately mapping sequences to their atomic coordinates [18, 19, 20]², researchers recognized that deep neural networks could capture the underlying biophysical grammar of proteins directly from evolutionary data. This has spurred the rapid development of generative AI models capable of unconditional and conditional *de novo* protein design [16, 21, 26]⁸. Today, the field is largely driven by two distinct but increasingly intersecting generative families: Protein Language Models (PLMs), which adapt natural language processing (NLP) architectures to treat amino acids as biological "words," and Diffusion Models (along with Flow Matching), which model the continuous geometric perturbations of atomic coordinates or the discrete corruption of sequences over time [3, 12, 127]¹⁰.

This comprehensive survey systematically examines the theoretical mechanisms, architectural innovations, and practical applications of AI-driven protein design. Drawing upon an exhaustive corpus of 300 academic publications [150]², this manuscript categorizes the current literature into coherent methodological families. By exploring the nuances of PLMs, continuous and discrete diffusion, inverse folding, and sequence optimization algorithms, this report aims to provide a rigorous foundational understanding of how artificial intelligence is decoding and rewriting the language of life.

Methods

The landscape of AI-driven protein design can be organized into distinct methodological families based on their underlying mathematical frameworks and the modalities (sequence versus structure) they model. This section details the foundations and recent advancements in these core areas.

Machine Learning and Biological Foundations

The foundation of modern protein design rests upon breakthroughs in natural language processing and generative modeling. The Transformer architecture [1]², initially developed for machine translation, introduced the self-attention mechanism, allowing models to weigh the global context of an entire sequence simultaneously. This architecture scales exceptionally well, giving rise to Masked Language Models (like BERT) [2]² and Autoregressive Causal Models (like GPT) [3]². When applied to biological sequences, these models learn to capture the long-range epistatic interactions that dictate protein folding.

Simultaneously, the development of score-based generative modeling and Denoising Diffusion Probabilistic Models (DDPMs) [4, 5]² revolutionized continuous data generation. Diffusion models learn to reverse a Markovian corruption process governed by Stochastic Differential Equations (SDEs) [6, 7]². For structured protein data, preserving the physical symmetries of 3D

space is critical. Equivariant Graph Neural Networks (EGNNs) and Invariant Point Attention (IPA) mechanisms ensure that the neural network's predictions respect SE(3) equivariance—meaning that rotations or translations applied to the input structure yield equivalent rotations and translations in the output [12, 13, 14]². Recently, Flow Matching has emerged as an alternative to diffusion, utilizing ordinary differential equations (ODEs) and stochastic interpolants to directly regress vector fields between noise distributions and data manifolds, offering faster and more stable training for geometric protein generation [9, 10, 11]².

Protein Language Models: Masked and Autoregressive Paradigms

The application of NLP frameworks to biological data operates on a powerful analogy: if amino acids are letters and protein domains are words, then a polypeptide chain is a sentence governed by evolutionary syntax. PLMs leverage the vast, unannotated sequence data available in repositories like UniProt and Pfam to learn high-dimensional semantic representations [49, 133, 150]¹³.

Masked Language Models and Structural Emergence

Masked Language Models (MLMs) are trained by corrupting a percentage of input amino acid tokens and forcing the model to reconstruct the original sequence using bidirectional context [36, 37, 38]². Models such as the Evolutionary Scale Modeling (ESM) series (e.g., ESM-1b, ESM-2) and ProtTrans have scaled to billions of parameters, trained on hundreds of millions of sequences [36, 40, 41]².

A profound phenomenon observed in MLMs is "structural emergence." Even though these networks are trained exclusively on 1D sequences, their internal attention mechanisms spontaneously learn to encode 3D spatial proximity [36, 133]¹⁴. Because the masking objective forces the model to capture co-evolutionary dependencies—where a mutation in one residue necessitates a compensatory mutation in a spatially adjacent residue to maintain thermodynamic stability—the PLM's attention maps correlate highly with physical contact maps. This continuous, high-dimensional semantic space serves as a potent feature extractor. Embeddings from MLMs have become the default input for downstream predictive tasks, including zero-shot mutational effect prediction, sub-cellular localization, and enzyme function annotation, frequently outperforming alignment-based evolutionary methods [42, 52, 70]².

Autoregressive Language Models for Sequence Generation

While MLMs excel at representation, Autoregressive (Causal) Language Models are explicitly designed for generation. By training the network to predict the next amino acid given all preceding tokens, models such as ProtGPT2 [79]², ProGen [76, 77, 78]², and Evo [83]² have demonstrated the capacity to hallucinate entirely novel, biologically viable sequences. Autoregressive PLMs generate sequences that respect the complex syntax of protein families, maintaining secondary structures and functional motifs without relying on explicit structural templates. For instance, ProGen leverages conditional control tags—such as taxonomic domains or enzymatic classes—to steer generation toward specific biological functions [76, 78]². Recently, models like Evo have expanded this paradigm to the genomic scale, modeling entire DNA, RNA, and protein sequences concurrently to capture long-range regulatory

interactions [83]². However, autoregressive decoding inherently suffers from unidirectional context constraints and is prone to repetition collapse, wherein the model falls into pathological homopolymer loops that undermine structural viability [131]²⁰.

PLM Architecture	Objective	Key Strengths	Representative Models
Masked (MLM)	Reconstruct corrupted tokens via bidirectional context.	Deep representation, structural emergence, zero-shot fitness prediction.	ESM-2 [38], ProtTrans [40], ProteinBERT [42]
Autoregressive	Predict the next token sequentially (causal masking).	Unconditional generation, high diversity, sequence-first design.	ProtGPT2 [79], ProGen [76], Evo [83]

Multimodal and Sequence-Structure Co-Design

Proteins exhibit a strict sequence-structure duality. Recognizing the limitations of sequence-only or structure-only generation, recent advancements have shifted toward multimodal PLMs. A flagship example is ESM3, a multimodal generative language model that jointly reasons over sequence, structure, and functional annotations [111, 140]².

To integrate 3D geometry into a discrete transformer architecture, multimodal PLMs often employ Vector Quantized Variational Autoencoders (VQ-VAEs) to discretize protein structures into a vocabulary of geometric tokens [112, 130]². This allows the model to process sequences and structures as parallel, interleaved text streams. Consequently, ESM3 can perform sequence-structure co-design, motif scaffolding, and inverse folding simultaneously.

Alternative approaches, such as SaProt, introduce structure-aware vocabularies by integrating 3D tokens (e.g., derived from Foldseek) directly alongside amino acid tokens, enabling the PLM to inherently grasp spatial constraints during pre-training [43, 60]². Furthermore, energy-based joint models like CarbonNovo align sequence and structural representations using unified objective functions, drastically improving sequence recovery and designability metrics [118]².

Continuous Structural Diffusion and Flow Matching

While PLMs excel at capturing evolutionary grammar, they lack explicit, continuous geometric awareness. To design backbones with atomic precision—a necessity for designing binding interfaces or enzyme active sites—researchers rely heavily on Diffusion Models and Flow Matching operating in continuous 3D space [141, 142]².

Denoising Diffusion on Geometric Manifolds

Continuous diffusion models represent structure generation as the reversal of a stochastic differential equation (SDE). In the forward process, native atomic coordinates $\mathbf{x}_0$ are gradually corrupted into isotropic Gaussian noise $\mathbf{x}_T$ over time $t \in [0, T]$ according to:

$$d\mathbf{x} = f(\mathbf{x}, t)dt + g(t)dw$$

The generative reverse process employs an equivariant neural network, typically parameterized by θ , to approximate the score function $\nabla_{\mathbf{x}} \log p_t(\mathbf{x})$, enabling the simulation of the reverse SDE to denoise $\mathbf{x}_T$ back into a structured protein backbone [189, 190]². Models like FrameDiff and RFdiffusion represent proteins as local coordinate frames for each residue, consisting of rotations in $SO(3)$ and translations in $\mathbb{R}^3$ [141, 144]². The forward noise process is modeled distinctly for translations (using standard Brownian motion) and rotations (using Brownian motion on the $SO(3)$ manifold), ensuring that the geometric validity of the peptide bonds is preserved throughout the denoising trajectory [185, 186]².

Motif Scaffolding and Programmable Design

The flagship model of this era is RFdiffusion, which fine-tuned the highly successful RoseTTAFold architecture to perform structural denoising [141]². RFdiffusion established state-of-the-art results across numerous tasks, most notably *motif scaffolding*. In motif scaffolding, a critical functional geometry—such as a viral epitope binder or an enzyme catalytic triad—is provided as a fixed structural condition. The diffusion model is then tasked with hallucinating a stable protein backbone around this motif that structurally supports it in the required orientation [141, 151]².

Similarly, Chroma introduced a highly programmable diffusion framework. Chroma utilizes classifier-free guidance and a custom energy function to steer the generation process,

allowing users to constrain generation by symmetry (e.g., enforcing C_3 or icosahedral oligomeric states), dimensions, or physical properties without retraining the underlying model [142]².

Flow Matching Architectures

Recently, Flow Matching (FM) has emerged as a deterministic, simulation-free alternative to standard diffusion. Flow matching constructs a vector field $v_\theta(x, t)$ that directly transports probability mass from a prior p_0 to the data distribution p_1 . The training objective is defined by minimizing the expected loss against a target vector field $u_t(x)$:

$$\mathcal{L}_{\text{FM}} = \mathbb{E}_{t, p_t(x)} \|v_\theta(x, t) - u_t(x)\|^2$$

FM provides straighter inference trajectories, leading to faster sampling and highly stable training dynamics. This framework is heavily utilized in architectures like FoldFlow, EigenFold, and ProHiFlo to handle multi-chain and complex protein backbones [147, 148, 149, 160]².

All-Atom Generation and Side-Chain Packing

Early diffusion models predominantly focused on C_α backbones, delegating the placement of side-chain atoms to downstream packing algorithms. However, protein function is inextricably tied to the precise atomic chemistry of side-chains. Consequently, fully atomic continuous diffusion models have recently been developed. Furthermore, diffusion models operating specifically on torsional angles (e.g., DiffPack) provide rigorous solutions for flexible side-chain packing within a fixed backbone context [154, 155]².

Discrete Sequence Diffusion

Extending diffusion to the discrete vocabulary of amino acid sequences presents unique mathematical hurdles, as standard Gaussian noise cannot be straightforwardly applied to categorical tokens. To bridge this gap, researchers have developed Discrete Denoising Probabilistic Models (D3PMs) [196, 198]².

D3PMs and Evolutionary Scale Diffusion

Discrete diffusion defines the forward corruption process via a series of Markov transitions

governed by a transition matrix $\mathbf{Q}$. For an amino acid token at time t , the transition is given by:

$$q(x_t | x_{t-1}) = \text{Cat}(x_t; x_{t-1} \mathbf{Q}_t)$$

There are two primary strategies for defining $\mathbf{Q}$. The *uniform noise* approach allows any amino acid to randomly mutate into any other amino acid with equal probability, mimicking random evolutionary drift. Alternatively, the *absorbing state* approach uses a special [MASK] token; during the forward process, tokens are progressively masked until the sequence is entirely obscured. The reverse generation process then corresponds to order-agnostic, iterative unmasking, guided by the neural network [196, 203, 208]².

A prominent application of discrete diffusion is EvoDiff, a general-purpose framework trained on evolutionary-scale natural protein sequences. Unlike autoregressive PLMs that decode sequentially, EvoDiff decodes in parallel across the entire sequence [191, 192]². This sequence-only discrete diffusion formulation has the distinct advantage of generating intrinsically disordered regions (IDRs). Structure-based continuous diffusion models are heavily biased by the Protein Data Bank (PDB), which is dominated by stable, crystallizable, and

well-folded proteins. Consequently, structure-based models struggle to design flexible linkers or intrinsically disordered domains. Because EvoDiff operates solely in the sequence space, it can reliably generate functional IDRs, bridging a crucial gap in the design of signaling proteins and flexible biosensors [191, 192]².

Diffusion Paradigm	Mathematical Framework	Key Capabilities	Representative Models
Continuous Diffusion	SDEs, SE(3) Equivariance	Motif scaffolding, geometric constraint satisfaction.	RFdiffusion [141], Chroma [142], FrameDiff [144]
Flow Matching	ODEs, Stochastic Interpolants	Faster inference, stable multi-chain topology generation.	FoldFlow [148], EigenFold [149], ProHiFlo [160]
Discrete Diffusion	Markov Transition Matrices (D3PM)	Intrinsically Disordered Regions (IDRs), parallel decoding.	EvoDiff [192], DiGress [198], NOS [201]

Inverse Protein Folding Architectures

The generation of a stable backbone is only half the protein design equation; the backbone must be populated with an amino acid sequence that spontaneously folds into that target geometry. This challenge is formalized as the Inverse Protein Folding problem [223, 224]².

Graph-Based Message Passing

The current gold standard for inverse folding is ProteinMPNN [221]². ProteinMPNN represents the target backbone as a k -nearest neighbor (k -NN) spatial graph. Nodes represent residues, and edges are featurized with interatomic distances and dihedral angles. Using a message-passing neural network (MPNN), the model updates latent node and edge representations to autoregressively decode the sequence. On structurally held-out backbones, ProteinMPNN routinely achieves sequence recovery rates exceeding 50%, vastly outperforming physical energy functions like Rosetta (typically ~30%) [221, 230]². Other powerful inverse folders include ESM-IF1, which utilizes Geometric Vector Perceptrons (GVPs) combined with a sequence-to-sequence transformer [223, 229]², and PiFold, which improves computational efficiency through structure-aware graph transformer layers [224]². ESM-IF1 is notable for its use of data augmentation; it was trained not only on experimentally

resolved PDB structures but also on millions of AlphaFold2-predicted structures, yielding highly robust generalization to novel backbone topologies [223, 236]².

Multi-State and Dynamic Inverse Folding

A fundamental limitation of classical inverse folding is the assumption that proteins exist as static monoliths. In biological reality, many proteins—such as biosensors, allosteric enzymes, and GPCRs—are highly dynamic, existing in conformational ensembles and transitioning between multiple states to perform their functions [235, 239]². To address this, researchers have developed DynamicMPNN and multi-state inference modifications for ProteinMPNN. By averaging the logits of two or more independent single-state embeddings during the decoding phase, these models can generate sequences that remain stable across multiple disparate conformations. This capability is critical for engineering molecular switches, motors, and smart therapeutics that actuate in response to environmental triggers [235, 239]².

Fitness Optimization and Active Learning

While generative models excel at mapping the distribution of naturally occurring proteins $p(\mathbf{x})$, protein engineering is fundamentally an optimization problem: the goal is to discover sequences that maximize a specific fitness landscape $f(\mathbf{x})$ (e.g., thermostability, catalytic turnover, or binding affinity). Bridging the gap between likelihood-based generation and objective-based optimization relies heavily on advanced sampling and active learning paradigms [260, 264, 269]².

Generative Flow Networks (GFlowNets)

Standard Reinforcement Learning (RL) techniques, when applied to sequence design, frequently suffer from severe mode collapse. The policy network rapidly converges on a single sequence or a narrow family of sequences that maximize the reward, losing all structural diversity. This is disastrous in wet-lab scenarios where downstream toxicity, synthesis constraints, or immunogenicity might invalidate the entire cluster of candidates [260, 261]². Generative Flow Networks (GFlowNets) provide a robust solution by reframing sequence generation as a flow of probabilities through a directed acyclic graph. GFlowNets are trained using trajectory balance objectives to ensure that the probability of sampling a candidate sequence is directly proportional to its reward: $P(x) \propto R(x)$. Consequently, GFlowNets explore and cover all high-reward modes in the fitness landscape, acting as an automatic diversity regulator. This "structural hedging" guarantees that the generative pipeline outputs a highly diverse portfolio of functional candidates, vastly increasing the probability of experimental success [260, 262, 266]².

Bayesian Optimization in Embedding Spaces

For multi-round, lab-in-the-loop engineering, Bayesian Optimization (BO) is the premier sample-efficient strategy. In BO, a Gaussian Process (GP) acts as a surrogate model to map the fitness landscape, providing both a point estimate of fitness and an explicit metric of epistemic uncertainty [264, 265]².

Historically, BO struggled with the high dimensionality of protein sequences. Modern approaches (e.g., BOES) solve this by conducting Bayesian optimization directly within the latent semantic embedding space of a pre-trained PLM (such as ESM-2) [271, 272]². By leveraging an acquisition function—like Expected Improvement (EI) or Upper Confidence Bound (UCB)—the algorithm balances exploitation (sampling near known high-fitness sequences) with exploration (sampling highly uncertain, uncharacterized regions). This permits batched, robotic experimentation workflows that converge on highly optimized enzymes or therapeutics in a fraction of the time required by classical directed evolution [269, 273, 279]².

Specialized Architectures for Antibodies and Protein Complexes

Beyond monomeric design, AI architectures are increasingly deployed to model multiprotein complexes and execute functional targeting, particularly in immunology.

De Novo Antibody and Nanobody Design

Antibodies constitute a multi-billion-dollar therapeutic class. Their binding specificity is governed by six hypervariable Complementarity-Determining Regions (CDRs) [290, 299]². Designing antibodies *de novo* is notoriously difficult due to the scarcity of high-quality antigen-antibody complex data and the immense conformational flexibility of the CDR-H3 loop [288, 289]².

Recent advancements have tailored diffusion models specifically for the immunoglobulin fold. Models such as DiffAb, IgDiff, and Antigen-specific RFdiffusion perform joint sequence-structure diffusion, conditioning the generative process on the antigen epitope and the rigid antibody framework [288, 289, 298]². Advanced iterations employ "force-guided sampling," which integrates differentiable biophysical force fields directly into the diffusion reverse process to resolve steric clashes and optimize attractive interactions, mitigating the lack of training data [287]². Furthermore, retrieval-augmented diffusion architectures (e.g., RADAb) incorporate structurally homologous motifs from existing databases as weak conditions, significantly improving the stability and viability of the generated CDRs [293]².

Protein-Ligand and Complex Modeling

In small-molecule drug discovery, modeling the dynamic interactions within binding pockets is essential. Traditional docking relies on rigid-receptor approximations, which fail to capture induced-fit conformational changes. Modern generative models, such as DiffDock and PackDock, cast docking as a generative diffusion problem [171, 174]². These models simultaneously simulate the continuous translation, rotation, and torsional flexibility of the ligand, alongside the side-chain flexibility of the protein pocket, yielding highly accurate holo-state predictions. Flow matching methodologies have also been extended to generate target-aware binders entirely from scratch, seamlessly uniting the discovery of small molecules and therapeutic peptides under a single mathematical framework [175, 176, 178]².

Challenges and Prospects

Despite the meteoric rise of AI-driven protein design, substantial challenges remain that must be addressed to fulfill the promise of autonomous, programmable biology.

Data Bias and the "Dark Matter" of the Protein Universe: Deep generative models are inherently bounded by the data upon which they are trained. The Protein Data Bank (PDB) suffers from severe selection bias; it is overrepresented by stable, soluble, and easily crystallizable proteins, while underrepresenting membrane proteins, highly dynamic complexes, and intrinsically disordered regions [140, 157]². Similarly, sequence databases like UniProt contain millions of "enzyme dark matter" sequences with entirely unknown catalytic functions. Generative models trained on this biased distribution often struggle to design proteins for extreme environments (e.g., high-temperature or highly acidic industrial conditions) without extensive fine-tuning.

Out-of-Distribution Generalization and "Hallucinations": When prompted to design proteins highly divergent from natural topologies, continuous diffusion models and autoregressive PLMs frequently experience out-of-distribution (OOD) failures. They may generate structures with severe steric clashes, chemically impossible bond angles, or fragmented geometries that score highly in the model's internal latent space but fail utterly under biophysical reality [159, 161]². Addressing this requires a stronger integration of physical inductive biases into the neural network architecture, either through explicit force-field guidance or by expanding training sets with synthetic, physics-simulated molecular dynamics (MD) trajectories [167, 168, 169]².

Closing the Gap Between In Silico Metrics and In Vitro Success: There currently exists a significant disconnect between *in silico* validation metrics (e.g., designability, sequence recovery, self-consistency RMSD) and true *in vitro* experimental success. A sequence may achieve perfect folding confidence (pLDDT) and sub-angstrom RMSD in AlphaFold2 verification, yet fail to express, aggregate into inclusion bodies, or exhibit zero binding affinity in the wet lab [256, 273]². The path forward relies on closed-loop, automated Design-Build-Test-Learn (DBTL) platforms. By tightly coupling generative AI output with high-throughput microfluidics, robotic synthesis, and automated affinity screening, researchers can generate vast quantities of negative and positive interaction data. Feeding this empirical data back into the models via Reinforcement Learning from Experimental Feedback (RLEF) will fundamentally align the generative objective with true physical function [269, 274, 284]².

Conclusion

The intersection of deep generative artificial intelligence and structural biology has inaugurated a new era of protein engineering. Through the robust semantic representations of Protein Language Models and the atomic precision of Diffusion and Flow Matching architectures, the field has moved beyond merely analyzing existing biological molecules to actively synthesizing novel solutions for pressing medical and industrial challenges. From the *de novo* design of highly specific antibodies to the engineering of functional enzymes and biosensors, the methodological toolkit—spanning continuous structural modeling, discrete sequence diffusion, and Bayesian optimization—has achieved unprecedented maturity.

As the domain progresses, overcoming the limitations of dataset bias, enhancing out-of-distribution generalization, and deeply integrating automated wet-lab feedback loops will be paramount. Ultimately, the synthesis of these AI-driven technologies promises to

transform protein design from a bespoke, trial-and-error craft into a highly predictable, programmable engineering discipline, fundamentally reshaping the future of therapeutics and synthetic biology.

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